Practical 1

a) Create stacked and grouped bar charts for comparison.

b) Build treemaps and pie charts for part-to-whole analysis.

c)Manage metadata (rename, hide, change data types).

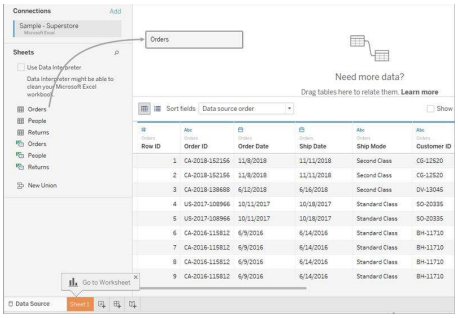
Step 1: Open Tableau and connect the dataset

1. Open Tableau

Open Tableau Desktop or Tableau Public on your computer.

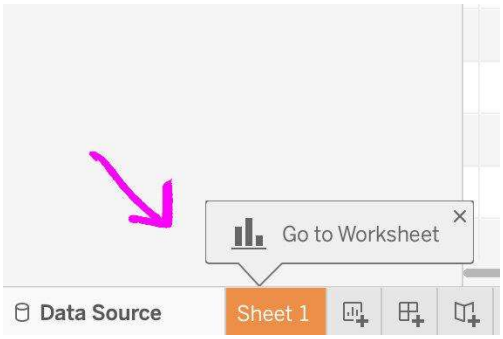
2. Connect to your data

On the left side, under Connect, click Text File if you have a CSV file, or Microsoft Excel if you have an Excel file.



3. Select Sample Superstore

Find and open your Sample Superstore.csv or Sample - Superstore.xlsx file. Wait for the data preview to load.

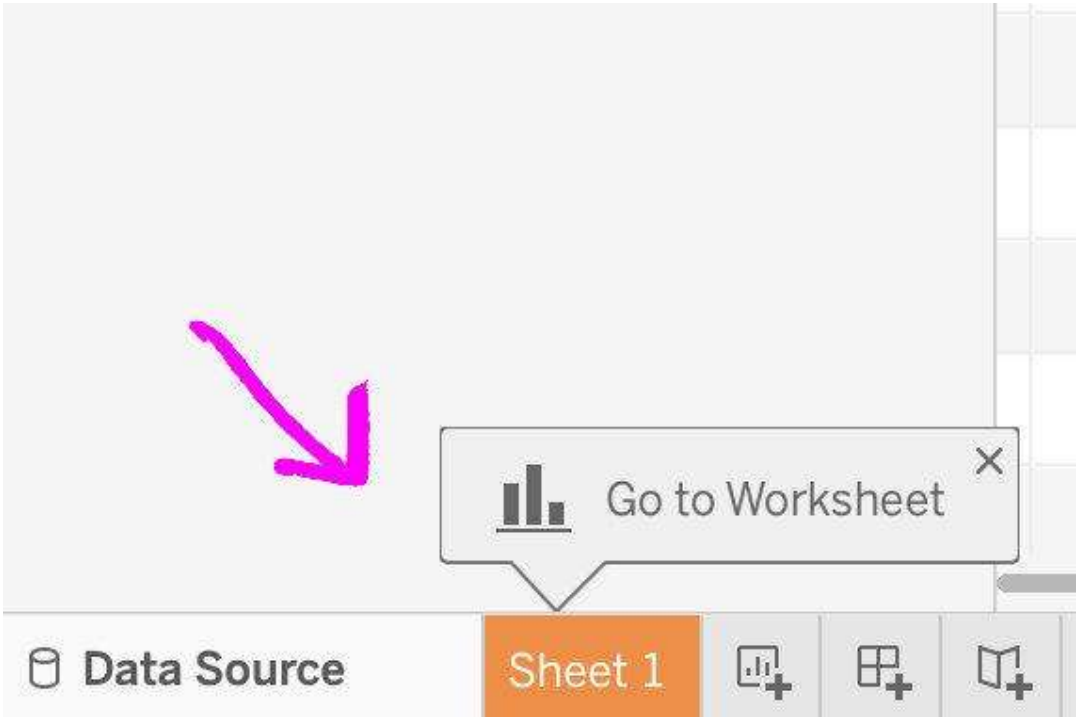


4. Open a worksheet

Click Sheet 1 at the bottom of the screen. You should see the Data pane on the left, with fields such as Category, Region, Sales, and Profit.

Practical 1: Question 1(a): Stacked Bar Chart in Tableau

Step 1 — Open a new worksheet



1. Look at the bottom of your Tableau window.

2. Click the New Worksheet icon (usually a small sheet icon with a plus sign), beside the existing sheet tabs.

3. A blank worksheet should open.

Step 2 — Build the stacked bar chart

Your question asks for stacked and grouped bar charts for comparison. We'll make the stacked bar chart first.

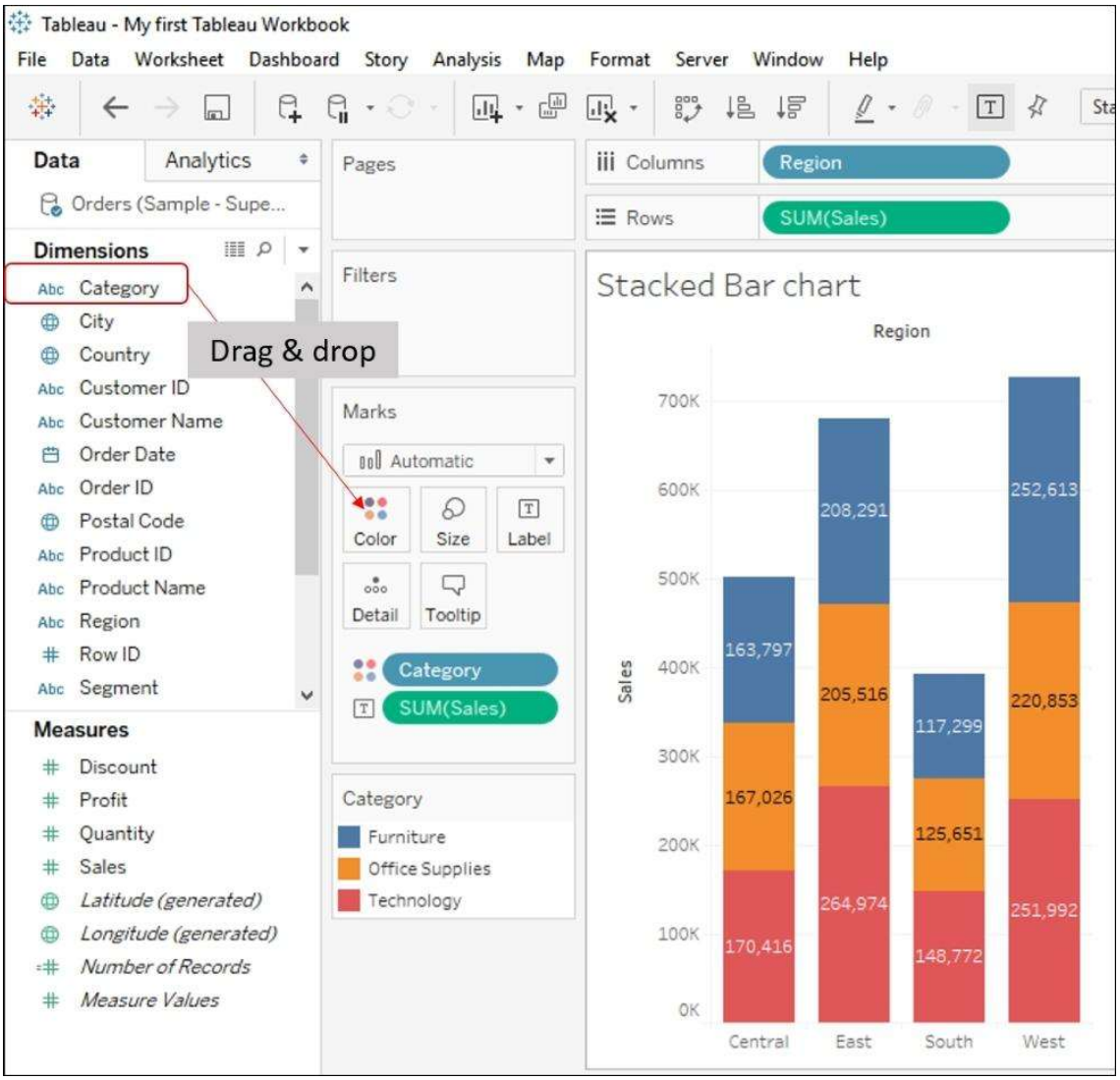
A. Drag Category to Columns

Find Category in the left-hand Data pane. Drag it onto the Columns shelf at the top.

B. Drag Sales to Rows

Find Sales in the Data pane and drag it onto the Rows shelf.

You should see bars for Furniture, Office Supplies and Technology.



C. Drag Region to Color

Find Region in the Data pane. Drag it onto Color on the Marks card.

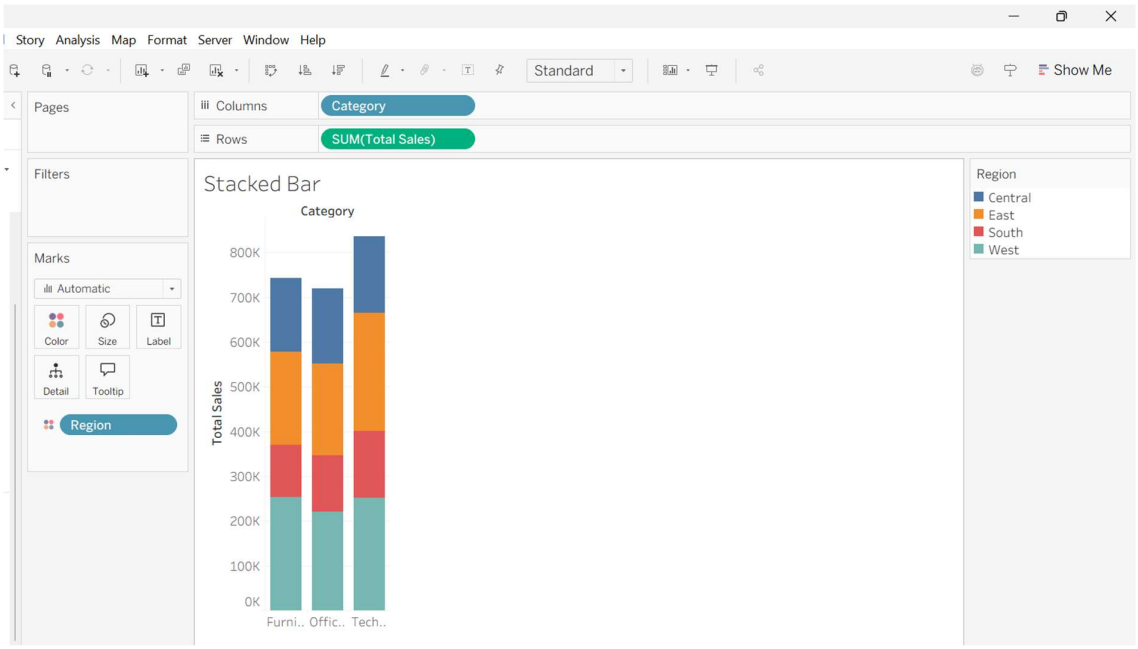
Each bar should now be divided into colored sections representing regions.

Step 3 — Make sure the bars are stacked

1. On the top menu, click Analysis.
2. Hover over Stack Marks.
3. Select On.

If your bars are already stacked, leave them as they are.

Step 4 — Check your result



You should have three bars — one for each product category — with colored sections for the different regions.

Step 5 — Create a new worksheet

1. At the bottom of Tableau, click the New Worksheet icon.
2. A blank worksheet should open.
3. We'll create a separate chart so you keep your stacked bar chart too.

Step 6 — Arrange the fields

A. Drag Category to Columns

Find Category in the left Data pane and drag it onto Columns.

B. Drag Region to Columns

Drag Region onto Columns, placing it immediately to the right of Category.

Your Columns shelf should show Category, then Region.

C. Drag Sales to Rows

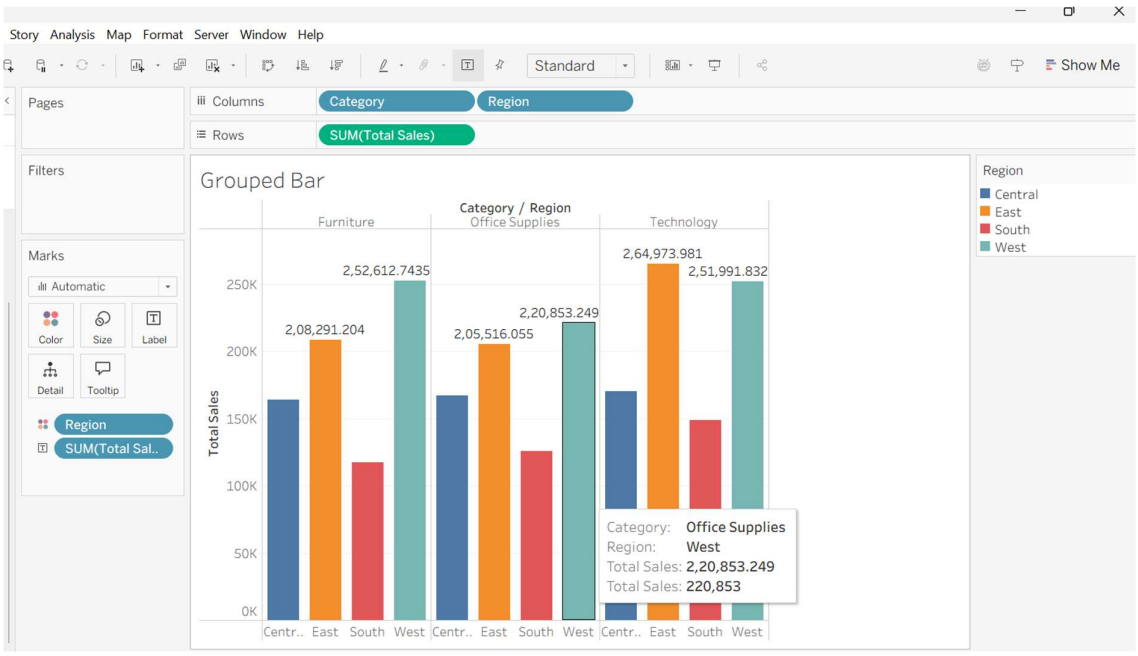
Find Sales in the Data pane and drag it onto Rows.

Step 7 — Select Bar chart

1. Find the Marks card.
2. Open its chart-type dropdown (it may say Automatic).

3. Select Bar.

Step 8 — Check your result



Your chart should show:

- Furniture — separate bars for the regions.
- Office Supplies — separate bars for the regions.
- Technology — separate bars for the regions.

b) Build treemaps and pie charts for part-to-whole analysis.

Part 1: Create a Treemap

Step 1 — Open a new worksheet

1. Look at the tabs along the bottom of Tableau.
2. Click the New Worksheet icon (usually a small sheet with a plus sign).
3. You should see a blank worksheet with your data fields on the left and the Show Me panel on the right.
4. Select Treemap in Show Me

Step 2 — Add Category, Sales, and Profit

1. Drag Category to Label

- Find Category in the Data pane on the left.

- Drag it onto Label in the Marks card.

2. Drag Sales to Size

- Find Sales in the Data pane.
- Drag it onto Size in the Marks card.

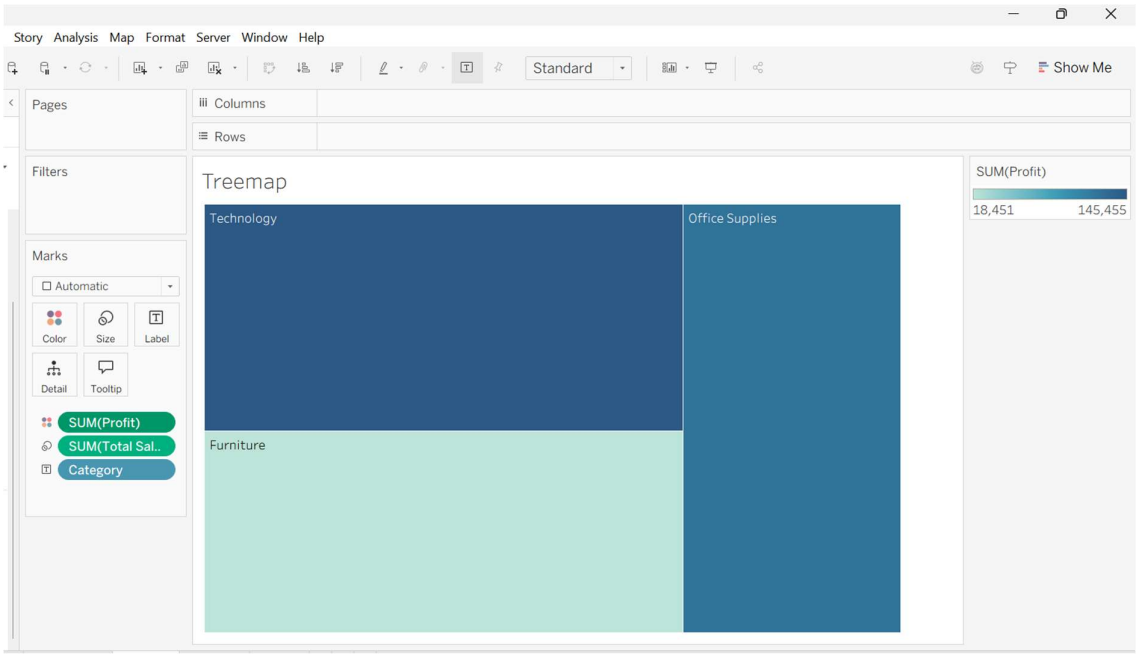
The rectangles should vary in size according to sales.

3. Drag Profit to Color

- Find Profit in the Data pane.
- Drag it onto Color in the Marks card.

The rectangles should now have colors representing profit.

Step 3 — Check your result



You should see a treemap made of rectangles, with Category labels, sizes based on Sales, and colors based on Profit.

Pie chart — Step 1: Open a new worksheet

At the bottom of Tableau, click the New Worksheet icon (the small sheet icon with a plus sign), or click the new worksheet button beside your existing sheet tabs.

Step 2: Select the Pie Chart

1. On the right side of Tableau, find the Show Me panel. Click it if it is closed.
2. Look for the Pie Chart icon in the chart options.
3. Click the Pie Chart icon.

Step 3: Add Category to Color

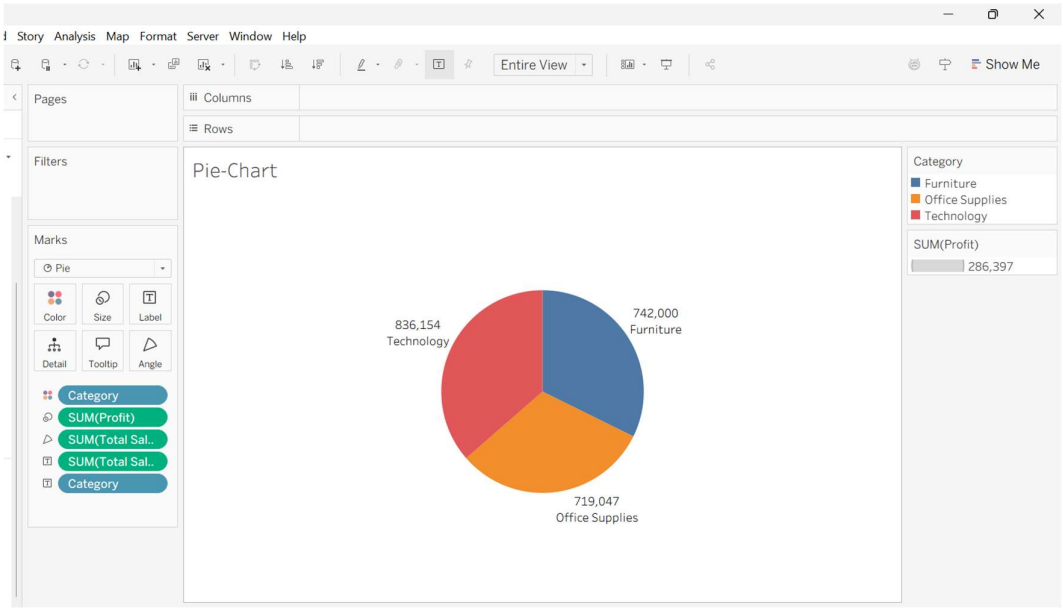
1. Look at the Data pane on the left.
2. Find Category.
3. Click and drag Category onto Color in the Marks card.

Step 4: Drag Sales to Angle

1. In the Data pane on the left, find Sales under Measures.
2. Click and drag Sales onto Angle in the Marks card.
3. Release it when Angle highlights.

Step 5: Add Category labels

1. On the Marks card, find Label.
2. From the Data pane on the left, drag Category onto Label on the Marks card.
3. If the category names don't appear, click Label and tick Show mark labels.



Expected result: The three slices will have different sizes based on their sales. A category with higher sales will have a larger slice.

c)Manage metadata (rename, hide, change data types).

Step 1: Rename a field

1. Open the Data pane on the left in Tableau.
2. Find the field named Sales under Measures.
3. Right-click Sales.
4. Select Rename.
5. Change its name to Total Sales and press Enter.

Step 2: Hide a field in Tableau

We'll hide Discount as an example. Your question bank asks you to demonstrate hiding a field, so this is enough to practise the operation.

1. Go to the Data pane on the left.
2. Find Discount under Measures.
3. Right-click Discount.
4. Click Hide.

That's it! The field should disappear from the visible list in the Data pane. This hides it from the worksheet field list; it does not delete the original data.

Step 3: Change the data type

We'll use Order Date as an example and make sure Tableau recognises it as a date.

1. Find Order Date

In the Data pane on the left, look for Order Date under Dimensions.

2. Right-click Order Date

A context menu will appear.

3. Select Change Data Type

Hover over or click Change Data Type to see the available types.

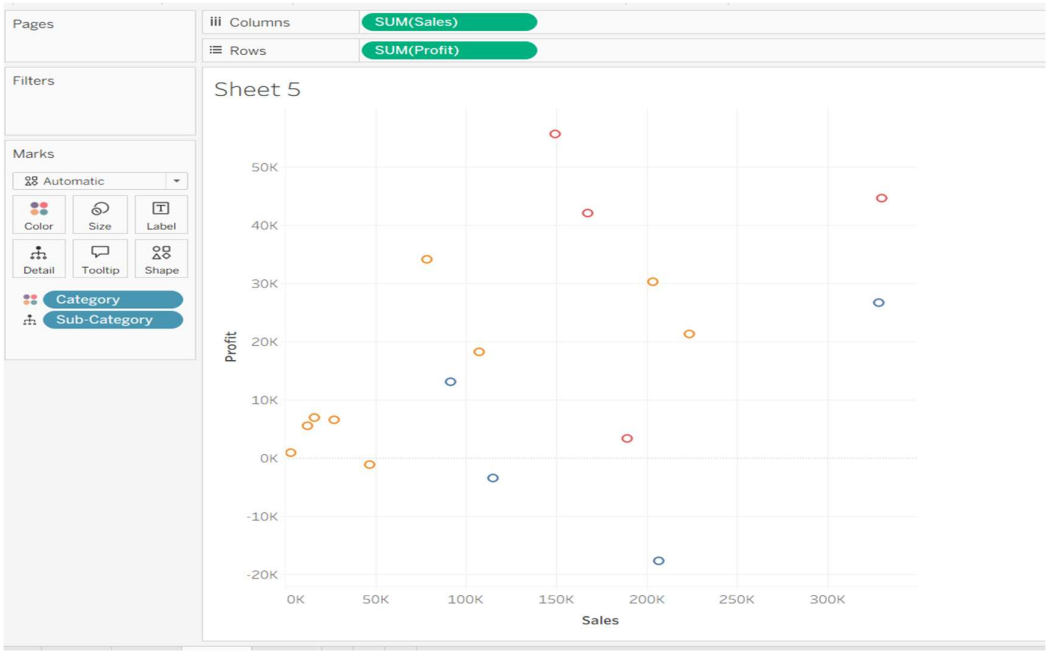
PRACTICAL 3.a) : Design scatter plots to identify relationships.

b) Create dual-axis and combination charts.

c) Implement different types of filters (dimension, measure, date).

3.a) Design scatter plots to identify relationships

1. Open a new worksheet.
2. Drag **Sales** to the **Columns** shelf.
3. Drag **Profit** to the **Rows** shelf.
4. Drag **Category** to **Color**.
5. Drag **Sub-Category** to **Detail** on the Marks card (this disaggregates the data points into individual marks).

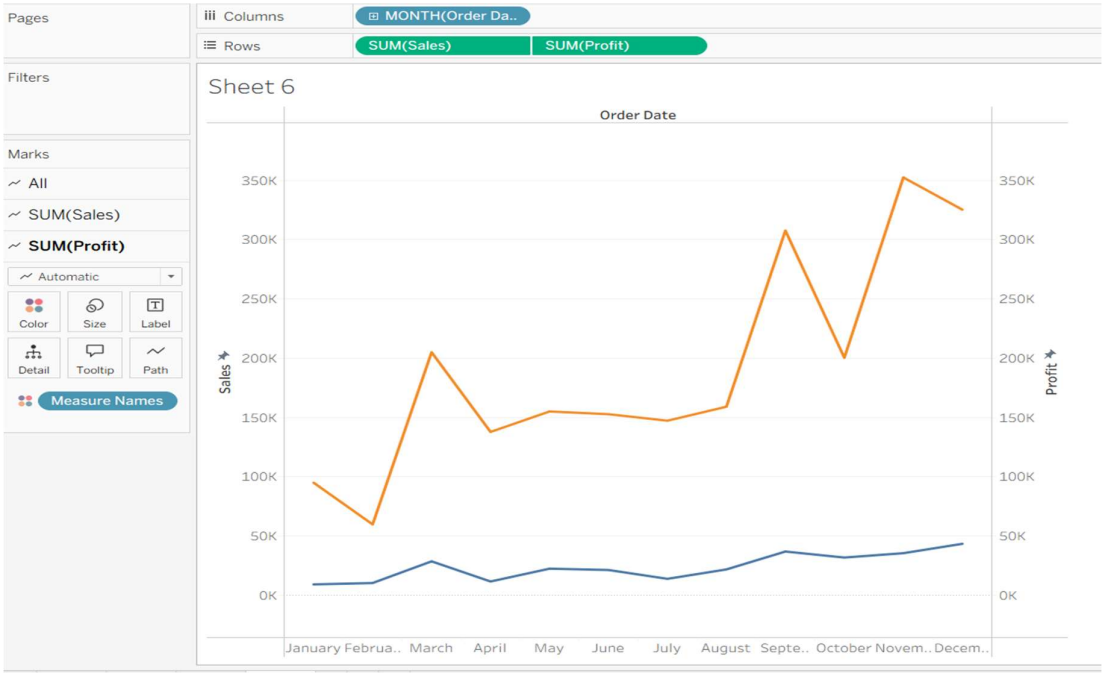


3.b) Create dual-axis and combination charts

Dual-Axis Chart:

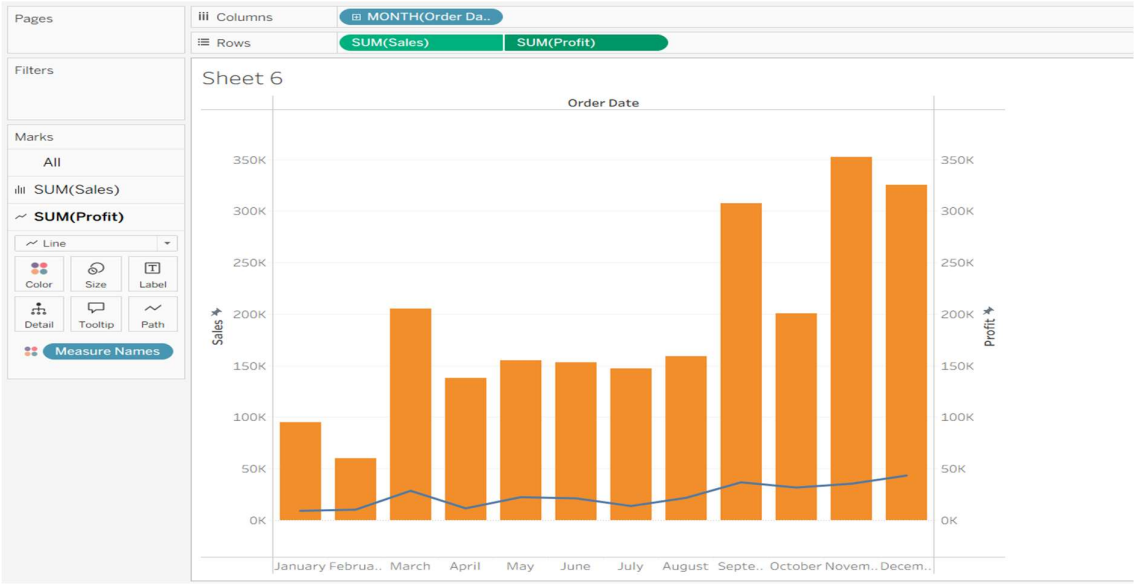
1. Drag **Order Date** to the **Columns** shelf (set to Year or Month).
2. Drag **Sales** to the **Rows** shelf.
3. Drag **Profit** and drop it next to SUM(Sales) on the **Rows** shelf.
4. On right side on Graph there is numbers click on it first and select → click **Synchronize Axis**.

5. Right-click the SUM(Profit) pill on the Rows shelf → select **Dual Axis**.



Combination Chart:

1. On the Marks card, click on the **SUM(Sales)** tab and set its Mark type to **Bar**.
2. Click on the **SUM(Profit)** tab and set its Mark type to **Line**.



3c. Set Up the Worksheet View First

1. **Build Base View**

- Drag **Sub-Category** to **Rows**.
- Drag **Sales** to **Columns**.

2. Dimension Filter

- Drag **Category** to **Filters**.
- Check **Office Supplies** (or any category) → click **OK**.
- Right-click the **Category** pill on Filters → select **Show Filter**.

3. Measure Filter

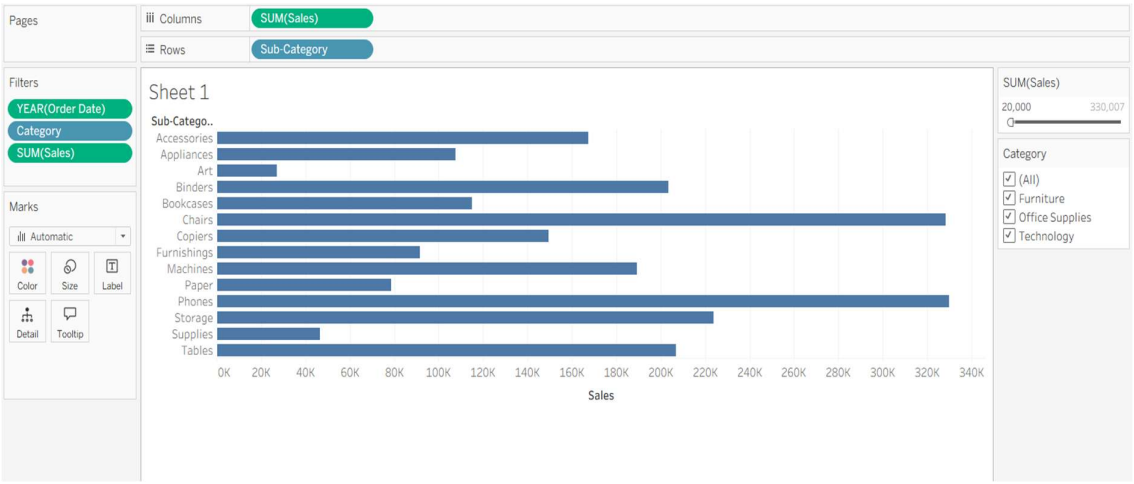
- Drag **Sales** to **Filters** → select **Sum** → click **Next**.
- Under **At least**, enter a value (e.g., 10000 or 20000) → click **OK**.
- Right-click the **SUM(Sales)** pill on Filters → select **Show Filter**.

4. Date Filter

- Drag **Order Date** to **Filters** → select **Years** → click **Next**.
- Check a year (e.g., **2016** or **(All)**) → click **OK**.
- Right-click the **YEAR(Order Date)** pill on Filters → select **Show Filter**

5.Adjust the SUM(Sales) Slider:

- On the right-hand **SUM(Sales)** filter slider, drag the slider handle slightly to the left (or click on the minimum value box and type 5000 or 10000) so more records meet the threshold.



Practical 4: 4.a) Create informative tooltips.

b) Implement "Viz in Tooltip".

>**Drag Category To Column**.

>**Drag Sales to Rows**.

>**Category To Color On Marks.**

> On the **Marks** card, click **Tooltip**.

> **In the text box, clear the existing default lines and type:**

Plaintext Formula:

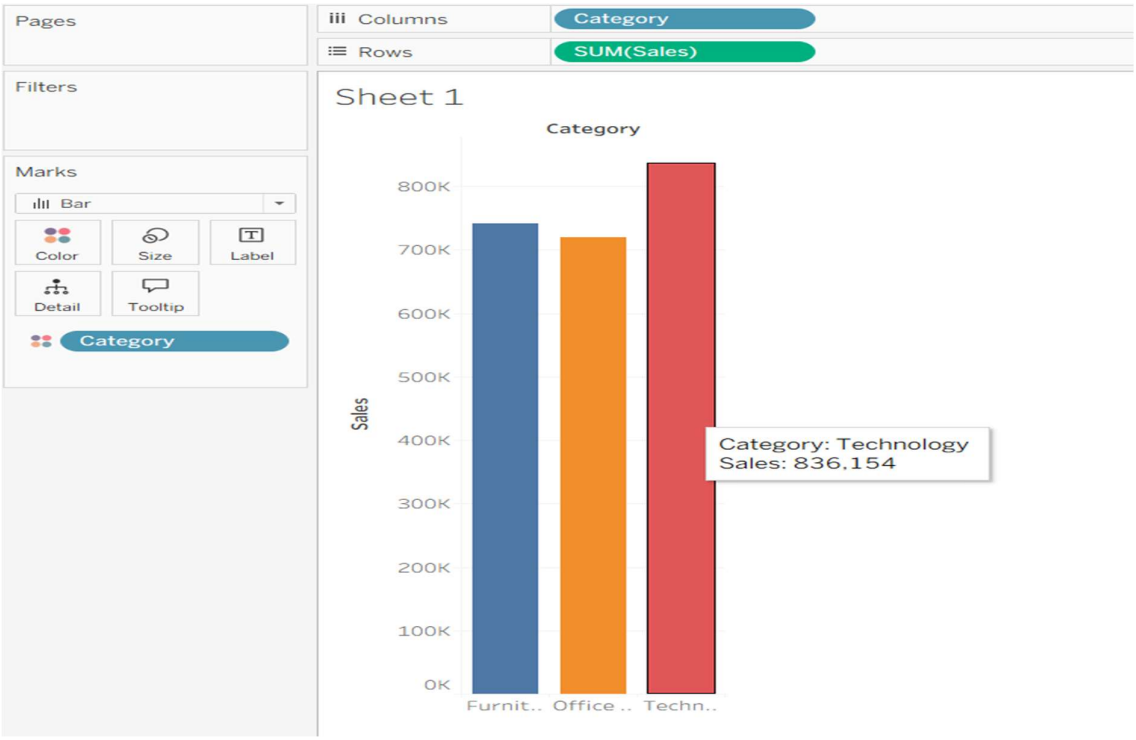
Sales Information

Category: <Category>

Sales: <SUM(Sales)>

> Click **OK**.

> Hover your mouse over any bar to verify the formatted tooltip appears.

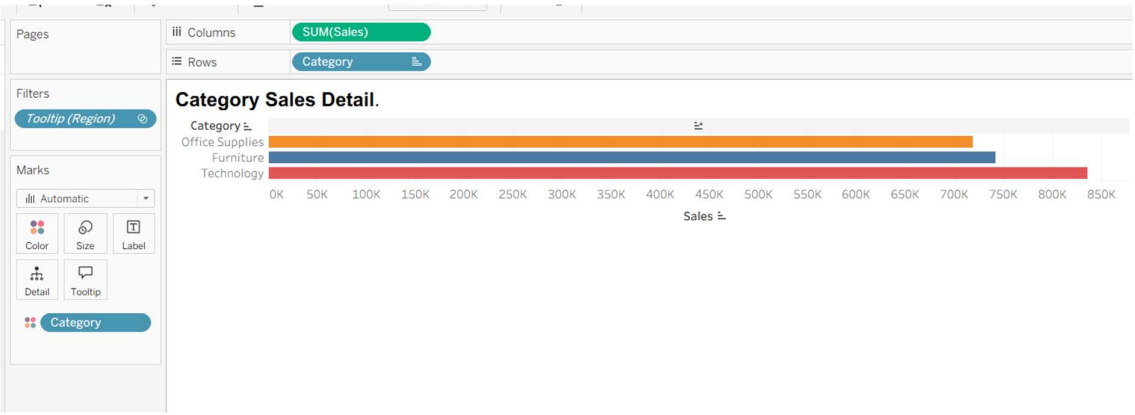


f) Implement Viz in Tooltip

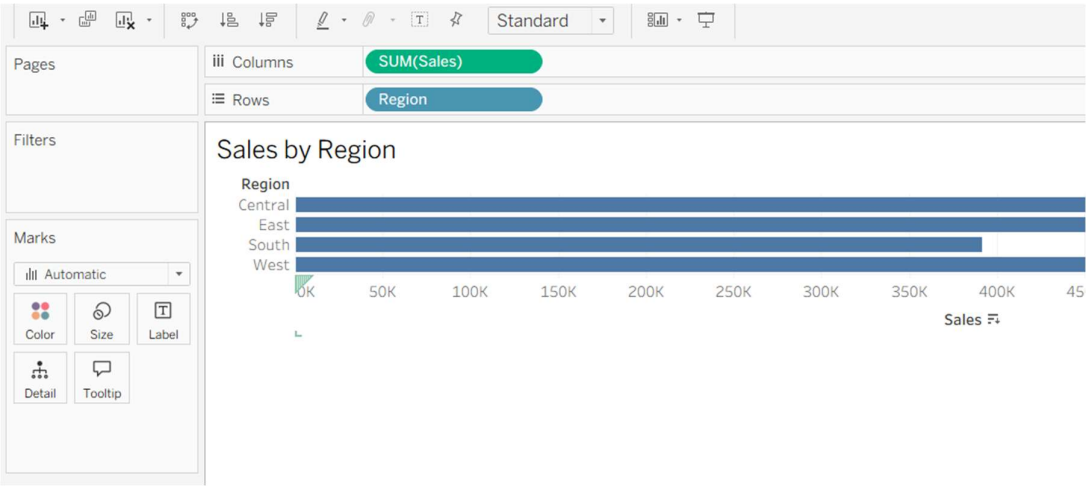
To complete the nested chart inside a tooltip:

1. **Create the Detail Sheet (Category Sales Detail):**

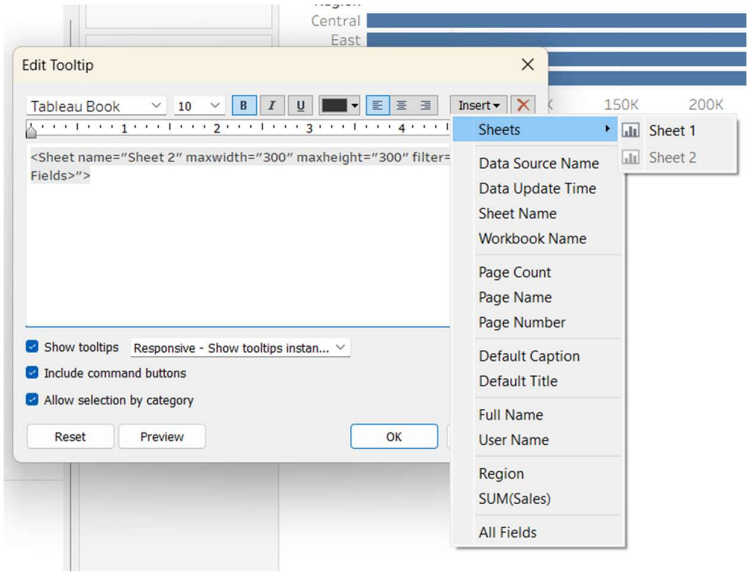
- Click the **New Worksheet** tab at the bottom.
- Rename the sheet to **Category Sales Detail**.
- Drag **Category** to **Rows**.
- Drag **Sales** to **Columns**.



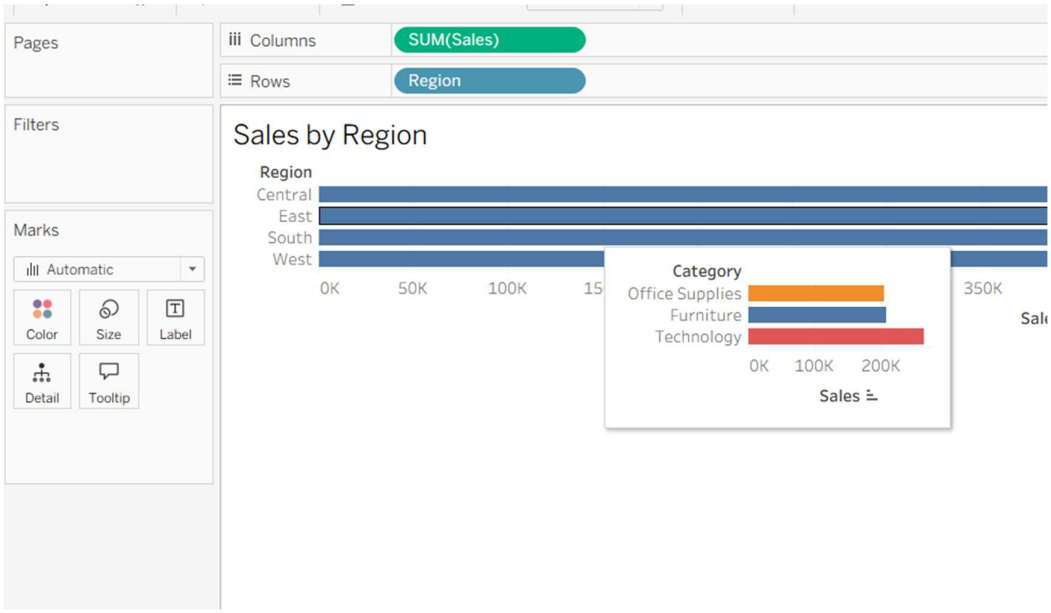
2. **Create the Main Sheet (Sales by Region):**
   - Click **New Worksheet** and rename it to **Sales by Region**.
   - Drag **Region** to **Rows**.
   - Drag **Sales** to **Columns**.



3. **Nest the Sheet into the Tooltip:**
   - On the **Sales by Region** sheet, click **Tooltip** on the **Marks** card.



- In the tooltip text editor, click **Insert** (top right) → hover over **Sheets** → **select sheet in which Category Sales Detail is there** .
- Click **OK**.



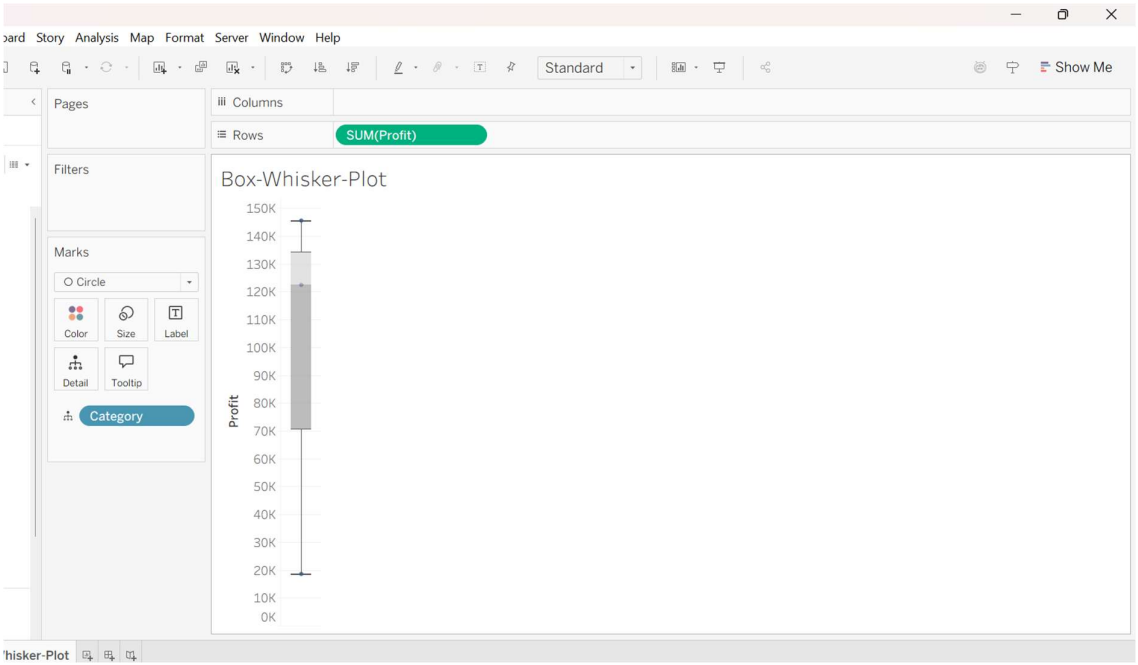
Practical 5

a) Build box-and-whisker plots.

Steps

1. Open a new worksheet.
2. Drag **Category** to **Columns**.
3. Drag **Profit** to **Rows**.
4. Click **Show Me → Box-and-Whisker Plot**.

Output



b) Develop histograms for distribution analysis.

Step 1: Open a new worksheet

1. Click the New Worksheet icon at the bottom of Tableau.
2. A blank worksheet will open.

Step 2: Find Quantity

1. Look at the Data pane on the left.
2. Find Quantity under Measures.

Step 3: Create the histogram

1. Click Quantity once in the Data pane.
2. Click Show Me in the top-right corner.
3. Click the Histogram chart icon.
4. Tableau should automatically create Quantity (bin) on Columns and CNT(Quantity) on Rows.

Step 4: Check the axes

Make sure your shelves show:

- Columns: Quantity (bin)
- Rows: CNT(Quantity)

Practical 9: Calculated Fields in Tableau

Dataset Used: Sample Superstore
Number of Records: 9,994

Dataset Fields

The following fields are used in this practical:

- Order Date
- Ship Date
- Category
- Sub-Category
- Region
- Segment
- Sales
- Profit
- Quantity

- Discount
- Customer Name
- State

Part A: Create Basic Calculated Fields Using Formulas

Aim

To create calculated fields in Tableau using mathematical formulas and use them to perform calculations on the Sample Superstore dataset.

Step-by-Step Procedure

Step 1: Open Tableau

1. Open Tableau Desktop or Tableau Public.
2. Open the Sample Superstore dataset.
3. Ensure that the data is loaded successfully.
4. Open a worksheet by clicking on Sheet 1.

Step 2: Locate the Data Pane

1. Look at the left side of the worksheet.
2. Locate the Data Pane, which contains dimensions and measures such as Sales, Profit, Quantity and Discount.
3. Right-click on an empty area inside the Data Pane.

Step 3: Create a Calculated Field

1. Select Create from the context menu.
2. Click on Calculated Field.
3. A calculation editor will appear on the screen.
4. Enter the required field name in the name box at the top.

Step 4: Create the Net Selling Price Field

1. Enter the calculated field name as Net Selling Price.
2. Type the following formula in the calculation editor:

[Sales] - [Sales] * [Discount]

3. Check whether the formula is valid. Tableau should display an indication that the calculation is valid.

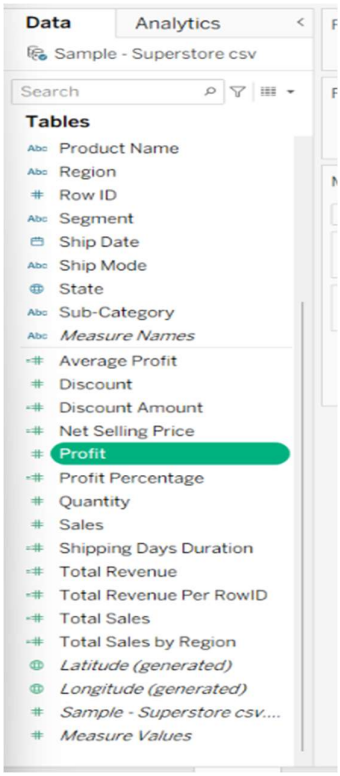
4. Click OK.

This formula calculates the selling price after subtracting the discount amount from Sales for each record.

Step 5: Create the Total Revenue Per RowID Field

1. Right-click inside the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name as Total Revenue Per RowID.
4. Enter the following formula:

[Net Selling Price] * [Quantity]



5. Check the formula for errors.
6. Click OK.

This formula multiplies the net selling price by the quantity to calculate the revenue value for each record.

Step 6: Verify the Calculated Fields

1. Look at the Data Pane.
2. Locate the newly created fields:

- Net Selling Price
- Total Revenue Per RowID

3. Drag Net Selling Price to the Rows shelf.
4. Drag Total Revenue Per RowID to the Rows shelf as well, or display them in a text table.
5. Verify that Tableau displays the calculated values.

Output

The calculated fields are successfully created and displayed in the Data Pane. They can now be used in worksheets for further analysis.

Part B: Perform Row-Level Calculations

Aim

To perform row-level calculations in Tableau using formulas that calculate values separately for each record in the dataset.

Step-by-Step Procedure

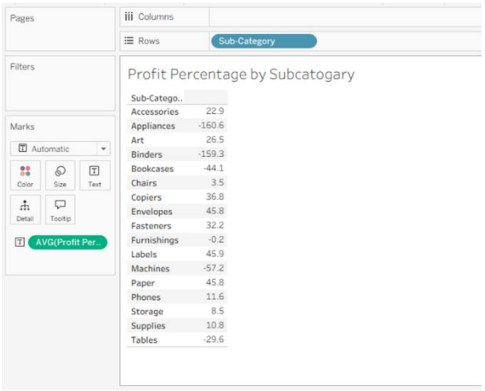
Step 1: Open the Worksheet

1. Open the Sample Superstore workbook.
2. Select a worksheet.
3. Locate the Data Pane on the left side.

Step 2: Create the Profit Percentage Field

1. Right-click inside the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name as Profit Percentage.
4. Enter the following formula:

([Profit] / [Net Selling Price]) * 100



5. Click OK after confirming that the formula is valid.

This formula calculates profit as a percentage of the net selling price for each record. If the net selling price is zero, the calculation may produce an undefined result; such cases should be handled separately if present.

Step 3: Create the Discount Amount Field

1. Right-click inside the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name as Discount Amount.
4. Type the following formula:

[Sales] * [Discount]

5. Click OK.

This formula calculates the discount amount for each record by multiplying Sales by Discount.

Step 4: Create the Shipping Days Duration Field

1. Right-click inside the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name as Shipping Days Duration.
4. Type the following formula:

DATEDIFF('day', [Order Date], [Ship Date])

5. Click OK.

This formula calculates the number of days between the order date and the ship date for each record.

Step 5: Display the Row-Level Calculations

1. Drag Order Date to the Rows shelf.
2. Drag Customer Name to the Rows shelf.
3. Drag Profit Percentage to the Text mark on the Marks card.
4. Similarly, create separate views to display Discount Amount and Shipping Days Duration.
5. To inspect individual records, include a unique order or row identifier if available, such as Row ID.
6. Check that each record displays its corresponding calculated value.

Output

The row-level calculated fields are successfully created. Tableau calculates profit percentage, discount amount and shipping duration separately for each record.

Part C: Perform Aggregate-Level Calculations

Aim

To create aggregate-level calculated fields in Tableau and analyse summarized values such as total sales, average profit and total revenue.

Step-by-Step Procedure

Step 1: Open a Worksheet

1. Open the Sample Superstore workbook.
2. Select a new worksheet.
3. Locate the Data Pane.

Step 2: Create the Total Sales Field

1. Right-click inside the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name as Total Sales.
4. Enter the following formula:

SUM([Sales])

5. Click OK.

This formula calculates the sum of sales values for the selected records.

Step 3: Create the Average Profit Field

1. Right-click inside the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name as Average Profit.
4. Type the following formula:

AVG([Profit])

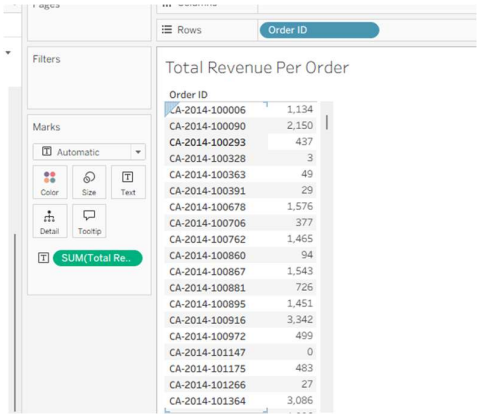
5. Click OK.

This formula calculates the average profit across the records included in the view.

Step 4: Create the Total Revenue Field

1. Right-click inside the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name as Total Revenue.
4. Type the following formula:

SUM([Total Revenue Per RowID])



5. Click OK.

This formula adds the values of Total Revenue Per RowID to calculate the total revenue for the selected records.

Step 5: Display the Aggregate Values

1. Open a new worksheet.
2. Drag Total Sales to the Text mark on the Marks card.
3. Observe the total sales value displayed in the worksheet.

4. Drag Average Profit to the Text mark to display the average profit.
5. Drag Total Revenue to the Text mark to display the total revenue.
6. Alternatively, place Region or Category on the Rows shelf to compare these aggregate values across different groups.
7. Ensure that Tableau displays the measures using the appropriate aggregation.

Step 6: Verify the Results

1. Check that Total Sales displays a summed value.
2. Check that Average Profit displays an average value.
3. Check that Total Revenue displays the sum of the calculated revenue values.
4. If a dimension such as Region or Category is added, verify that the aggregate values change according to the selected group.

Output

The aggregate-level calculated fields are successfully created. Tableau displays summarized values for total sales, average profit and total revenue. These values can be analysed across regions, categories and other dimensions.

Practical 10

a) Design dashboards for multiple devices.

Step 1: Create a dashboard

1. Click the New Dashboard icon at the bottom.
2. A blank dashboard opens.
3. In the left pane, under Size, select Desktop Browser (1000 × 800).

Step 2: Add worksheets

1. Under Sheets, drag Histogram onto the dashboard.
2. Drag Box-Whisker-Plot onto the dashboard.
3. Arrange them so both charts are visible.
4. Keep the charts Tiled for easier layout management.

Step 3: Show the dashboard title

1. In the left pane, tick Show dashboard title.

2. Double-click the title.
3. Enter Sales Analysis Dashboard.
4. Click OK.

Step 4: Open Device Preview

1. Click Device Preview in the left Dashboard pane.
2. The device preview toolbar appears at the top.
3. Find the Device type dropdown.

Your screenshot shows this toolbar, so you've already reached this stage.

Step 5: Preview the desktop layout

1. In Device type, select Desktop.
2. Check that both charts are visible and readable.
3. Adjust chart sizes if necessary.

Step 6: Preview the tablet layout

1. Open the Device type dropdown.
2. Select Tablet.
3. Click Add Tablet Layout if you want a separate layout for tablets.
4. Arrange the charts to fit the tablet screen.
5. Make sure both charts and their labels are readable.

Step 7: Create the phone layout

1. Open the Device type dropdown.
2. Select Phone.
3. Click Add Phone Layout if that button appears.
4. In the left pane, select the Phone layout.
5. Under Layout – Phone, choose Custom if you want to arrange the charts independently.
6. Stack Histogram and Box-Whisker-Plot vertically.
7. Select Fit width under Size – Phone so the dashboard fits the phone width.
8. Adjust the height if needed.

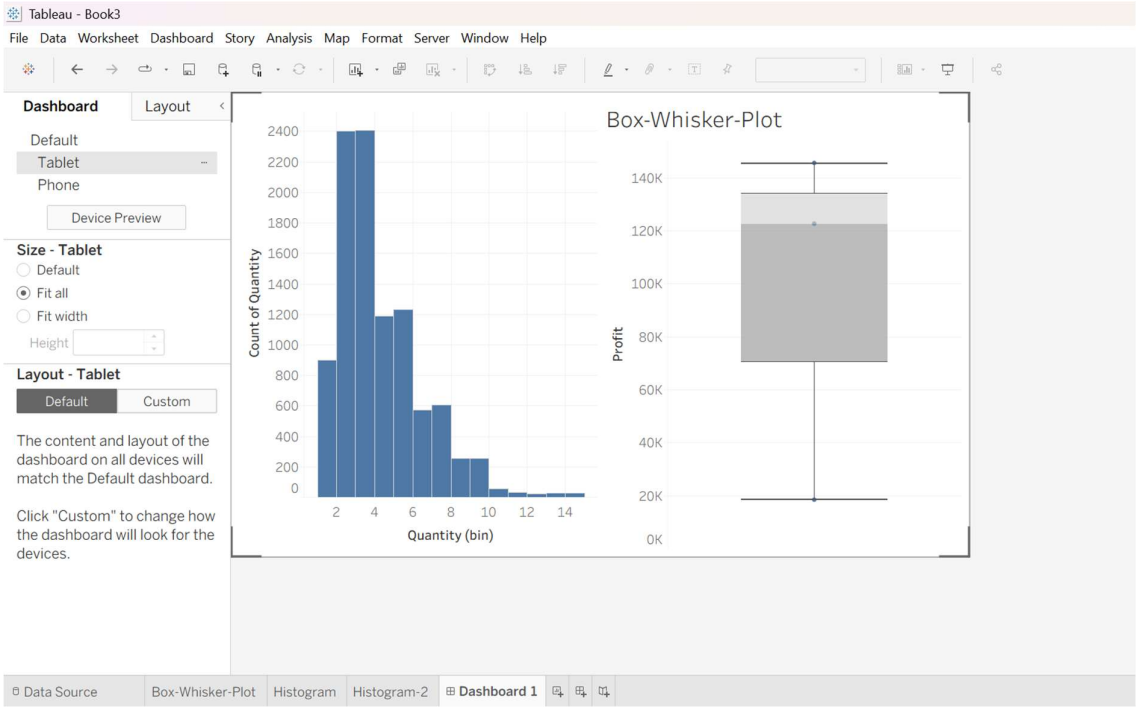
Your screenshot already shows a Phone layout with both charts stacked vertically and Fit width selected.

Step 8: Check each device

1. Select Desktop and check the desktop layout.
2. Select Tablet and check the tablet layout.
3. Select Phone and check the phone layout.
4. Ensure charts are not cut off and labels remain readable.

Expected output

A Tableau dashboard containing the Histogram and Box-and-Whisker Plot, with layouts that can be previewed and adjusted for desktop, tablet, and phone devices.



b) Build a complete data story using the Story feature.

Step 1: Create a new Story

1. Look at the bottom of Tableau, beside your worksheet and dashboard tabs.

2. Click the New Story icon — the small book-shaped icon with a plus sign.

3. A new Story page will open.

Step 2: Set the Story size

1. Look at the left-side Story pane.

2. Find Size.

3. Select Desktop Browser or a suitable desktop size if available.

Step 3: Add the first story point

1. Under Sheets, find Histogram.

2. Drag Histogram onto the blank Story canvas.

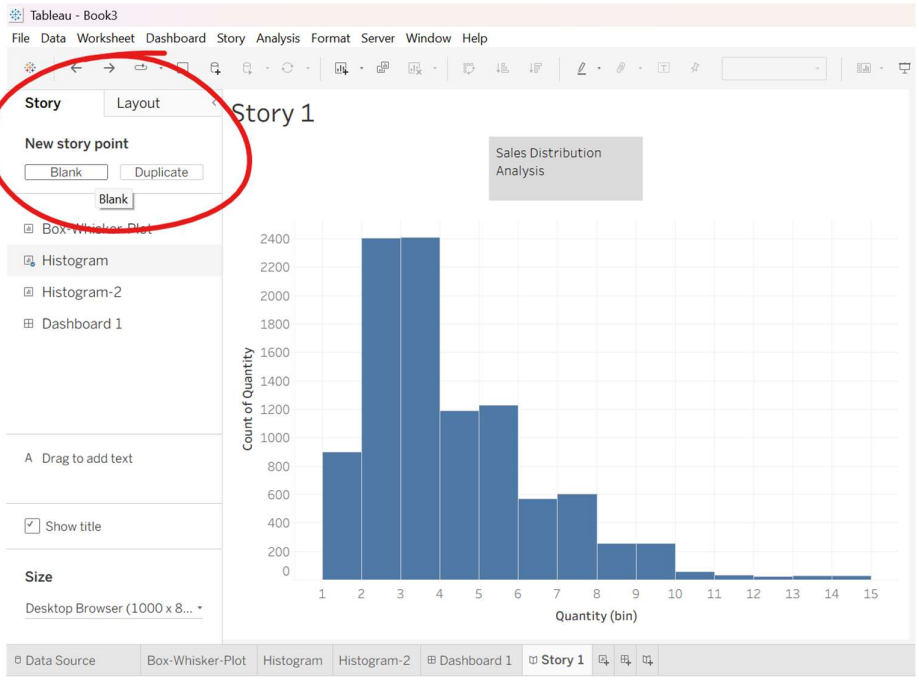
3. Tableau will create your first story point.

4. In the caption area, enter:

Sales Distribution Analysis

Step 4: Add the second story point

1. Click Blank or New Story Point at the top of the Story canvas.



2. Drag Box-Whisker-Plot from Sheets onto the canvas.

3. Enter the caption:

Profit Distribution Analysis

Step 5: Add the dashboard as a story point

1. Click Blank or the New Story Point button again.
2. Find Dashboard 1 under Sheets or the available dashboard items in the left pane.
3. Drag it onto the Story canvas.
4. Enter the caption:

Overall Data Analysis

Step 6: Add a description to each point

Use the caption or description area to explain what each view shows.

- Point 1 — Sales Distribution: Shows how frequently different quantity ranges occur in the dataset.
- Point 2 — Profit Distribution: Displays the spread of profit values using a box-and-whisker plot.
- Point 3 — Overall Analysis: Combines the charts into a dashboard for a broader view of the data.

Note: Your current histogram uses Quantity, and your box plot uses Profit, so these captions describe those charts accurately.

Step 7: Arrange the story points

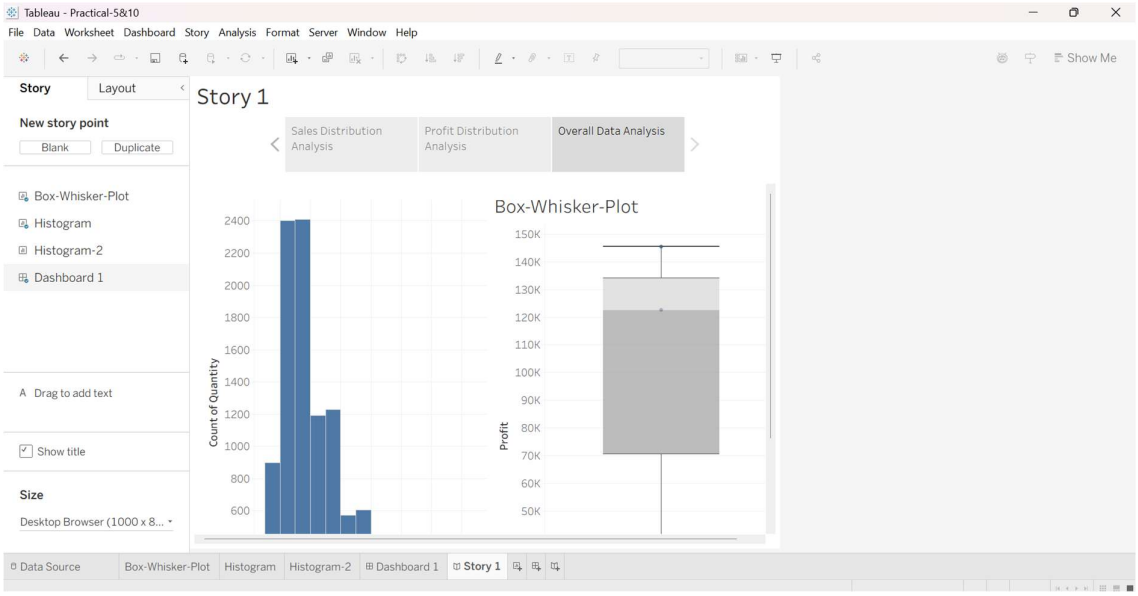
1. Check that the story points appear in this order:
    - Sales Distribution Analysis
    - Profit Distribution Analysis
    - Overall Data Analysis
2. Click each story point to confirm the correct view appears.
3. Make sure the captions are readable.

Step 8: Save the workbook

1. Click File → Save As.
2. Choose your preferred folder.
3. Enter Practical10_Data_Story.
4. Save as .twb or .twbx if you need to package the data with the workbook.

Expected output

A Tableau Story containing three sequential story points, using the Histogram, Box-and-Whisker Plot, and Dashboard to present a connected analysis of the data.



Practical 12: Trend Lines in Tableau

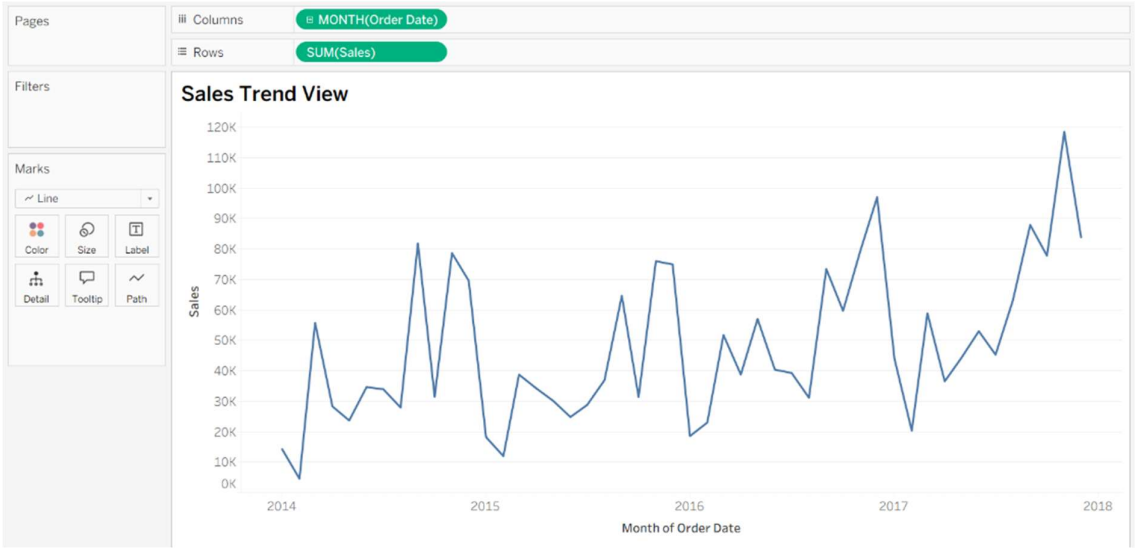
Dataset: Sample – Superstore

a) Add and Interpret Trend Lines (Linear and Exponential)

Procedure

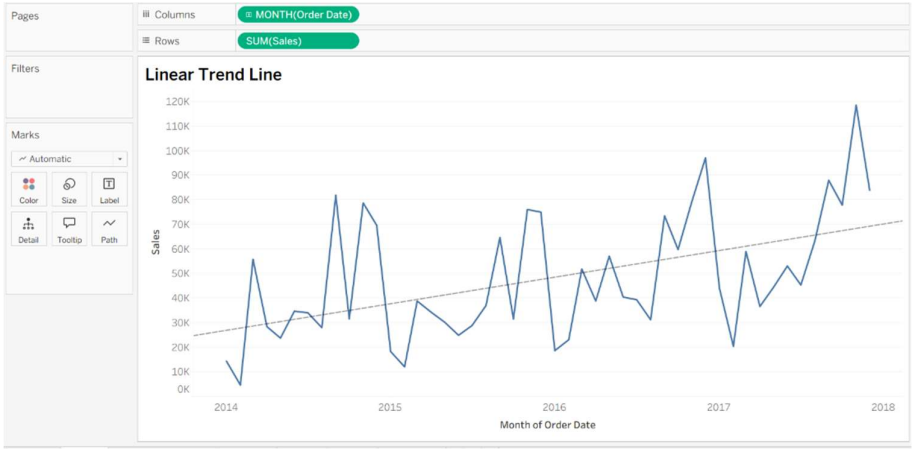
Step 1: Create Sales Trend View

1. Open Tableau and connect to the Sample – Superstore dataset.
2. Open a new worksheet and rename it Sales Trend View.
3. Drag Order Date to Columns and select Month.
4. Drag Sales to Rows.
5. Change the Marks type to Line.



Step 2: Add a Linear Trend Line

1. Open the Analytics pane.
2. Drag Trend Line onto the view and drop it on Linear.
3. Alternatively, select Analysis → Trend Lines → Show Trend Lines.
4. Observe the direction of the trend line.

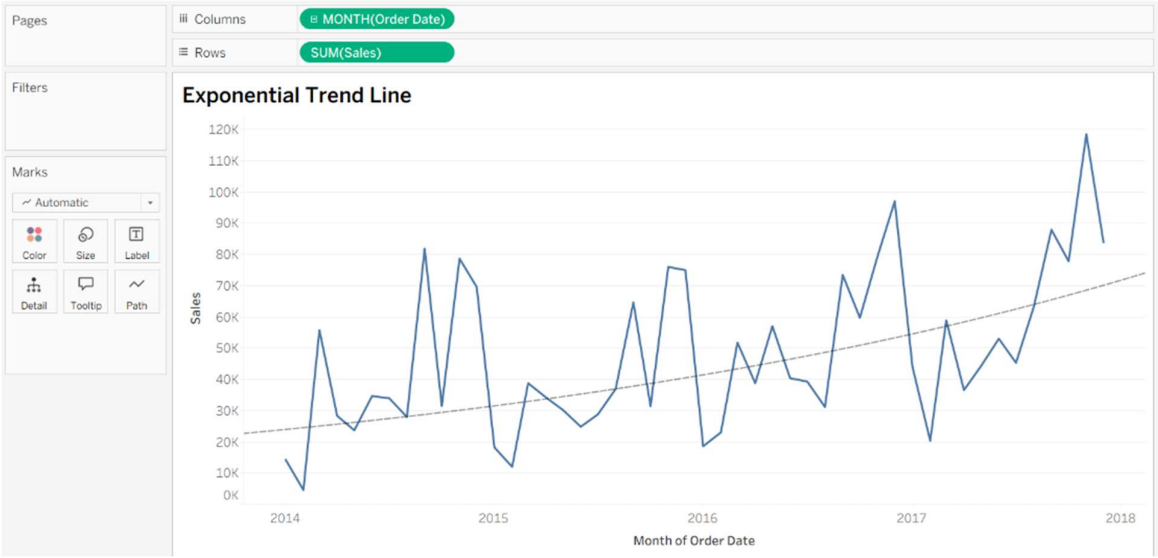


Output: Linear trend line added to the monthly Sales view.

Step 3: Add an Exponential Trend Line

1. Select Analysis → Trend Lines → Edit Trend Lines.
2. Select Exponential as the model type.
3. Click OK.

4. Observe the shape of the trend line.



Output: Exponential trend line added to the monthly Sales view, if supported by the data.

Observation:

- An upward-sloping trend line indicates an overall increase in Sales.
- A downward-sloping trend line indicates an overall decrease in Sales.
- A curved exponential line represents an exponential relationship in the data.

b) Customize and Analyze Trend Models

Procedure

Step 1: Create a Scatter Plot

1. Open a new worksheet and rename it Sales and Profit Trend Analysis.
2. Drag Sales to Columns.
3. Drag Profit to Rows.
4. Select Circle from the Marks card to create a scatter plot.
5. Drag Order ID to Detail on the Marks card.

Step 2: Add a Trend Line

1. Select Analysis → Trend Lines → Show Trend Lines.
2. Observe the trend line on the scatter plot.

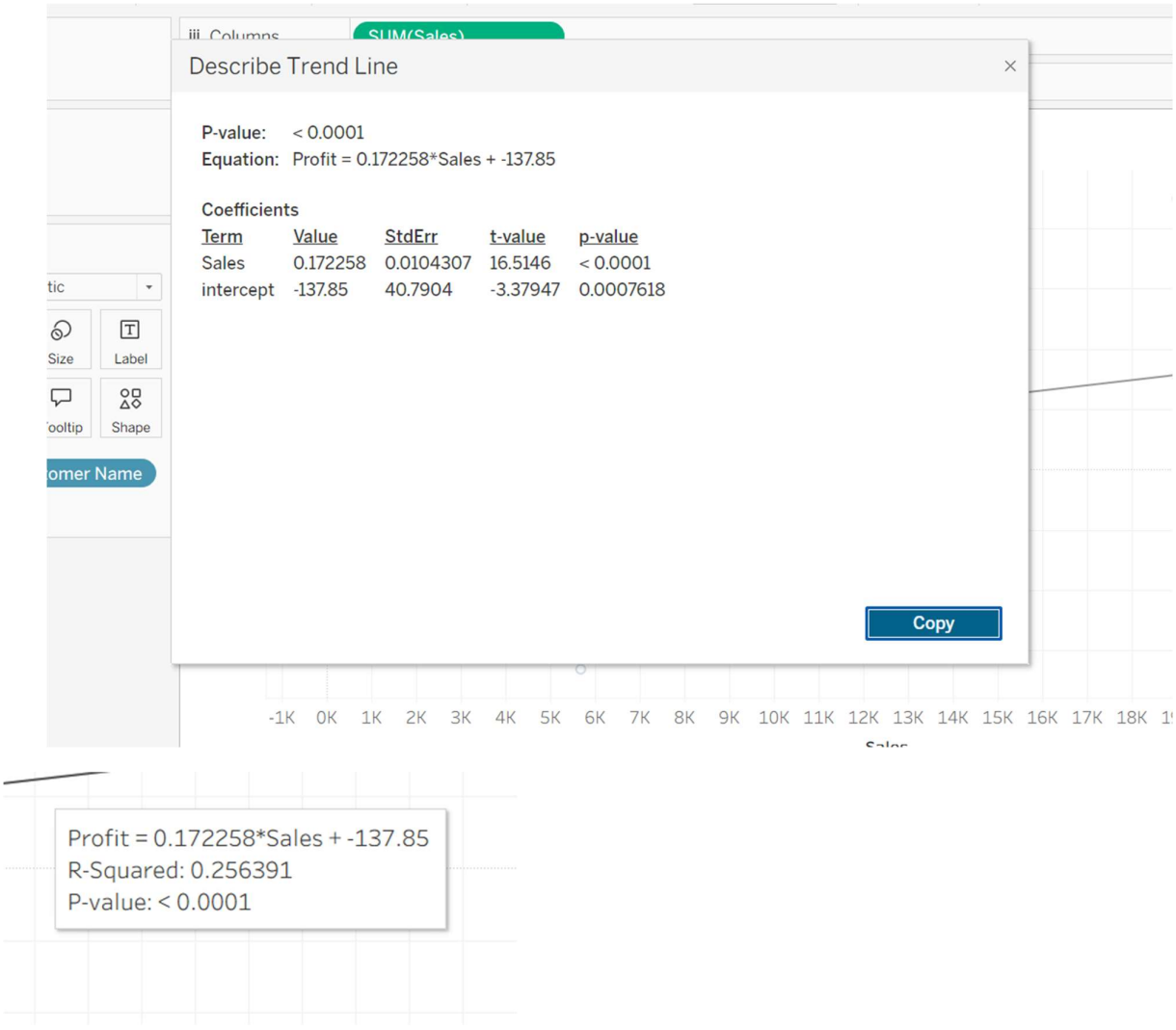
Output: Trend line added to the Sales and Profit scatter plot.

Step 3: Describe the Trend Line

1. Right-click the trend line.
2. Select Describe Trend Line.
3. Observe the model equation, R-squared value, P-value and other statistical details.
4. Close the dialog box.

Step 4: View the Trend Line Tooltip

1. Hover the mouse pointer over the trend line.
2. Observe the equation, R-squared value and P-value displayed in the tooltip.



Output: Trend line statistics are displayed for analysis.

Observation:

- The equation represents the fitted model.
- R-squared indicates how much variation is explained by the model.

- The P-value helps assess statistical significance.
- Coefficients, standard error and t-value provide additional statistical information.